

QDGset : A Large Scale Grasping Dataset generated with Quality-Diversity

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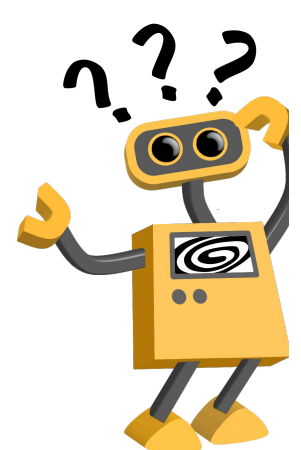


Abstract : QDGset is a framework for scaling up object-centric 6DoF grasp data generation. Using this framework, an extensible dataset of 62M grasps on 40K simulated objects has been created.

Introduction

Challenge of learning to grasp in robotics

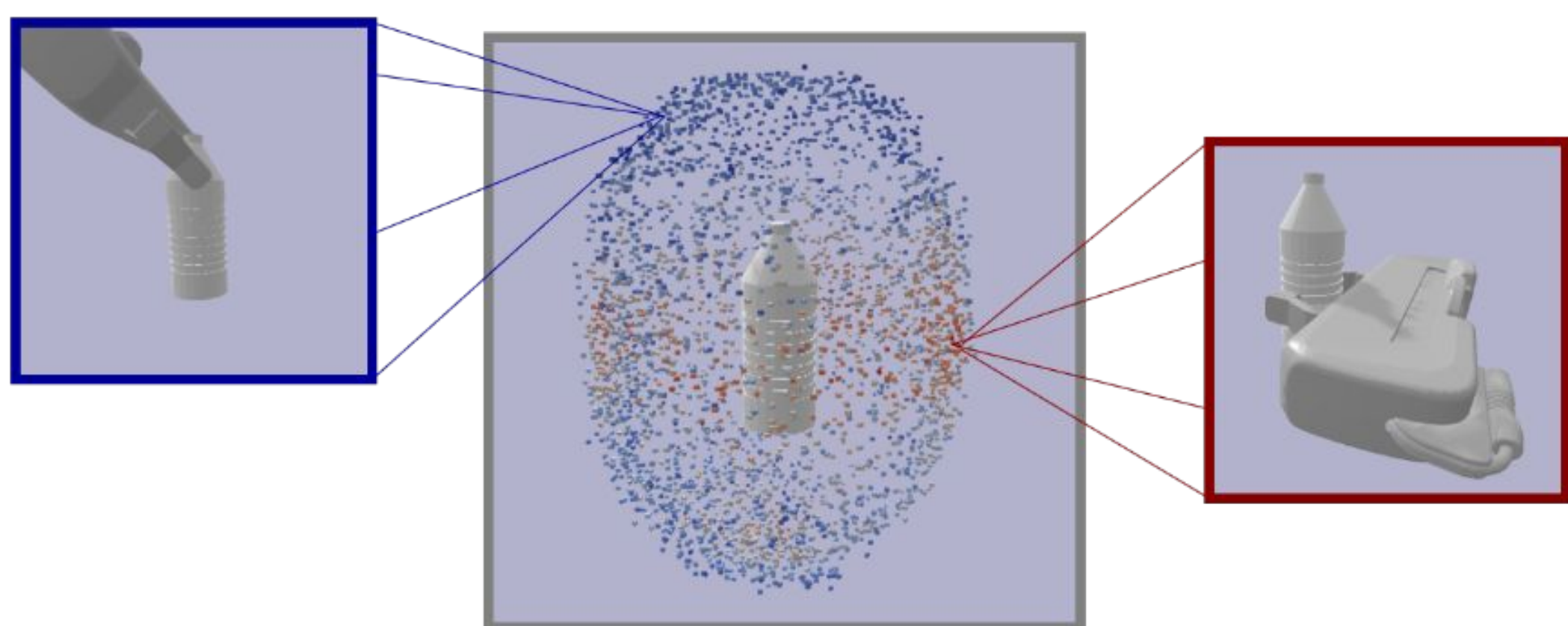
- Grasping $\Rightarrow$ sparse reward task
- Data-Driven methods require big demonstration dataset



How to automatically generate a grasp demonstrations dataset in a sparse reward environment ?

Problem Framing

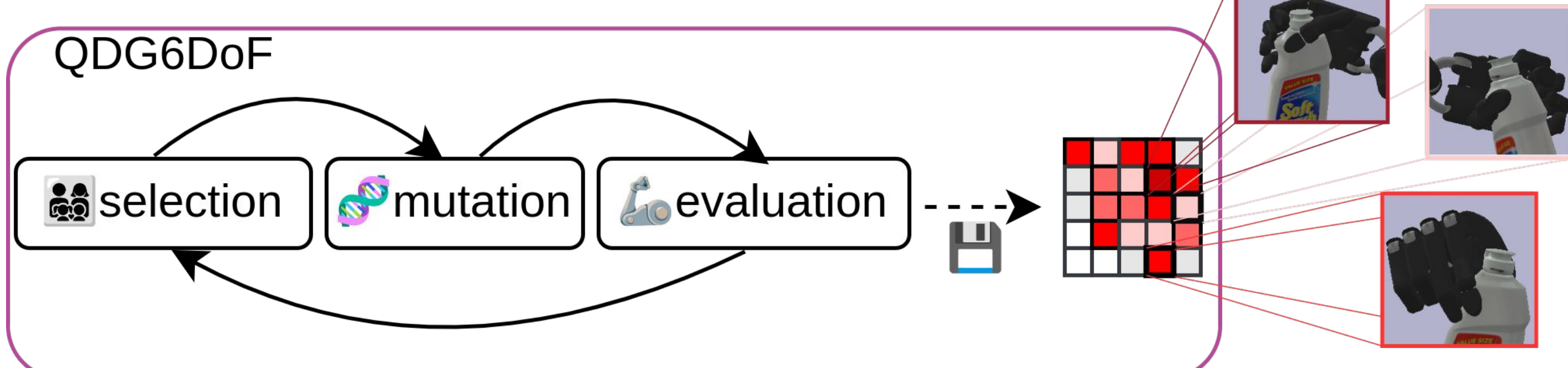
- Dataset of object-centric grasp demonstrations $g \in SE(3) \times \mathbb{R}^n$
- One demonstration = one 6-DoF end-effector grasp in the object reference frame [1][2]



Generating grasps with Quality-Diversity (QD)

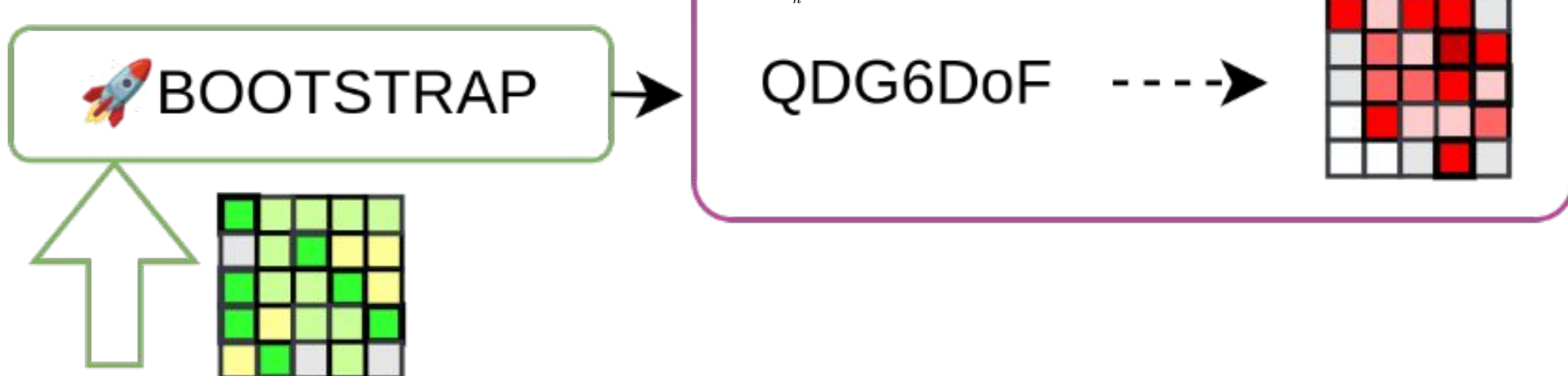
- $\left\{ \begin{array}{l} \forall b \in \mathcal{B}_{reach}, \exists \theta \in A, d_{\mathcal{B}}(\phi_{\mathcal{B}}(\theta), b) < \epsilon \text{ diverse solutions} \\ \forall \theta' \in A, \theta' = \operatorname{argmax}_{\theta \in N(b_{\theta'})} f(\theta) \text{ best-performing solutions} \end{array} \right.$

- Efficient exploration of sparse reward environments
- Evolutionary algorithm $\Rightarrow$ selection-mutation loop



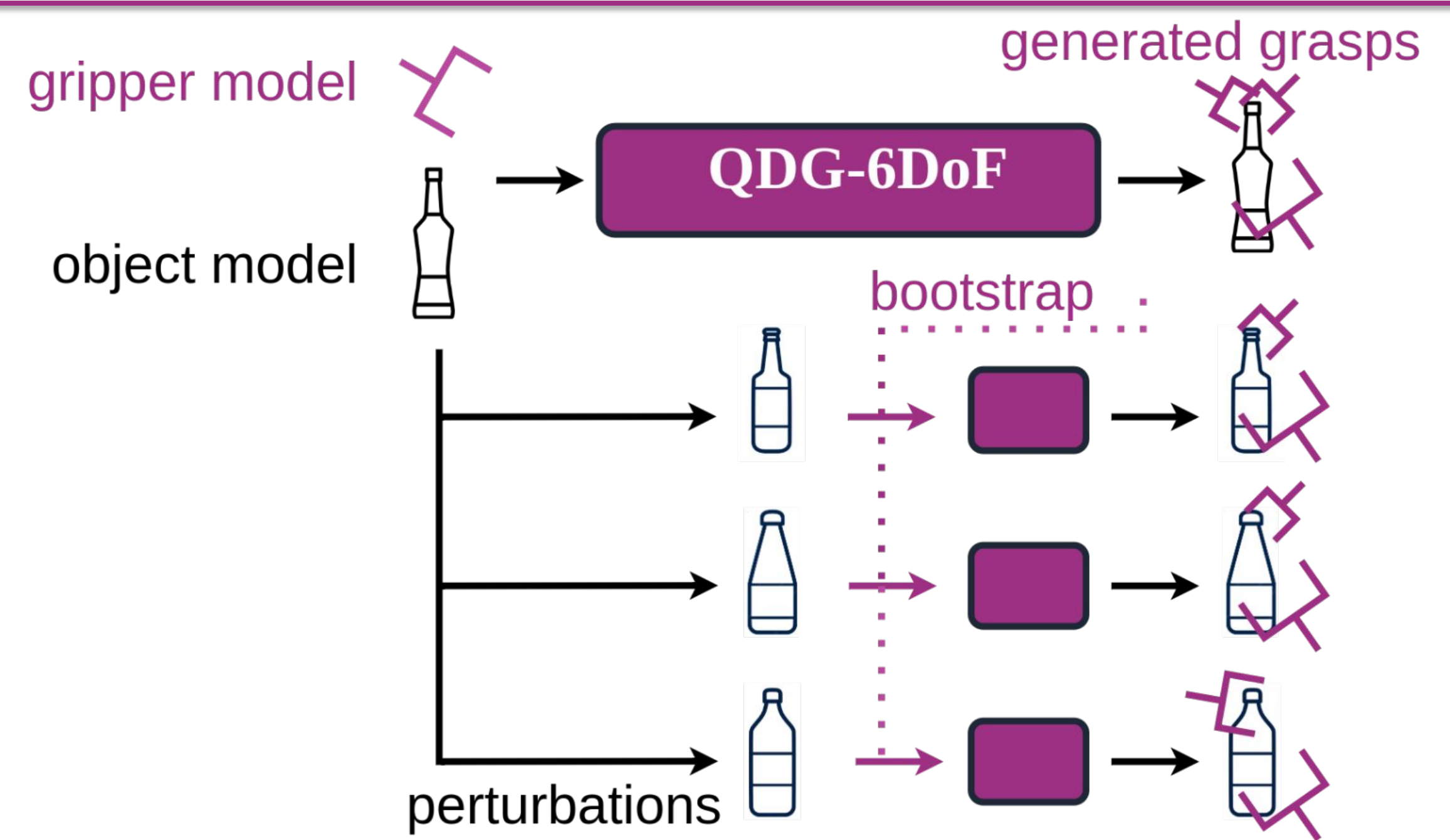
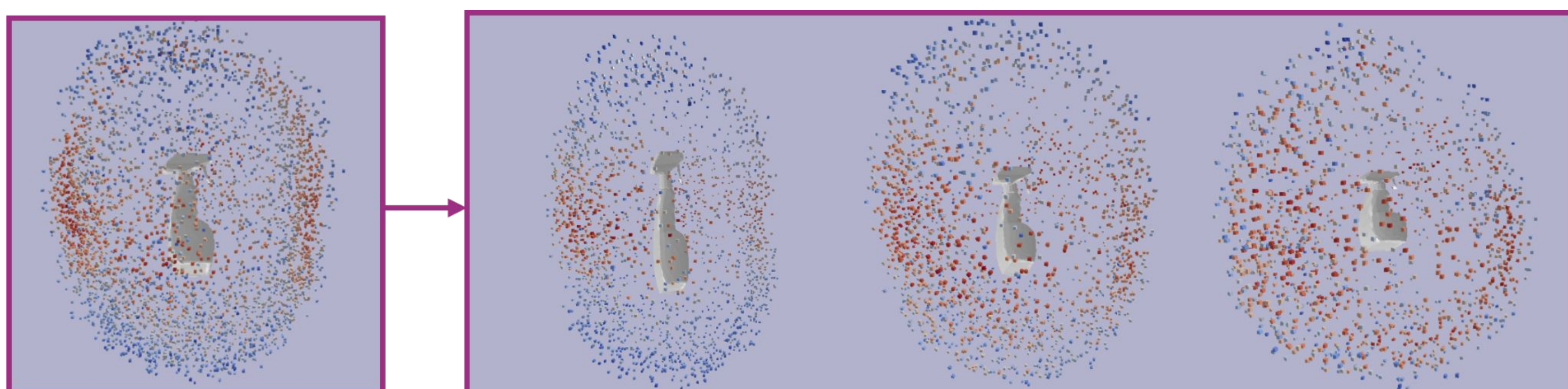
Methods

Scale up the production of synthetic grasping datasets 🚀



Bootstrap speeds up and scales up demonstrations

- Run QD optimisation on a given object A_s^b
- Use A_s^b to bootstrap the optimisation on similar objects



Object Data Augmentation

- Core initial object dataset P , $n = 29\,651$ objects

$$\mathcal{P}_i = \{p_j \in \mathbb{R}^3 \mid p_j = (x_j, y_j, z_j), j = 1, 2, \dots, m_i\}$$

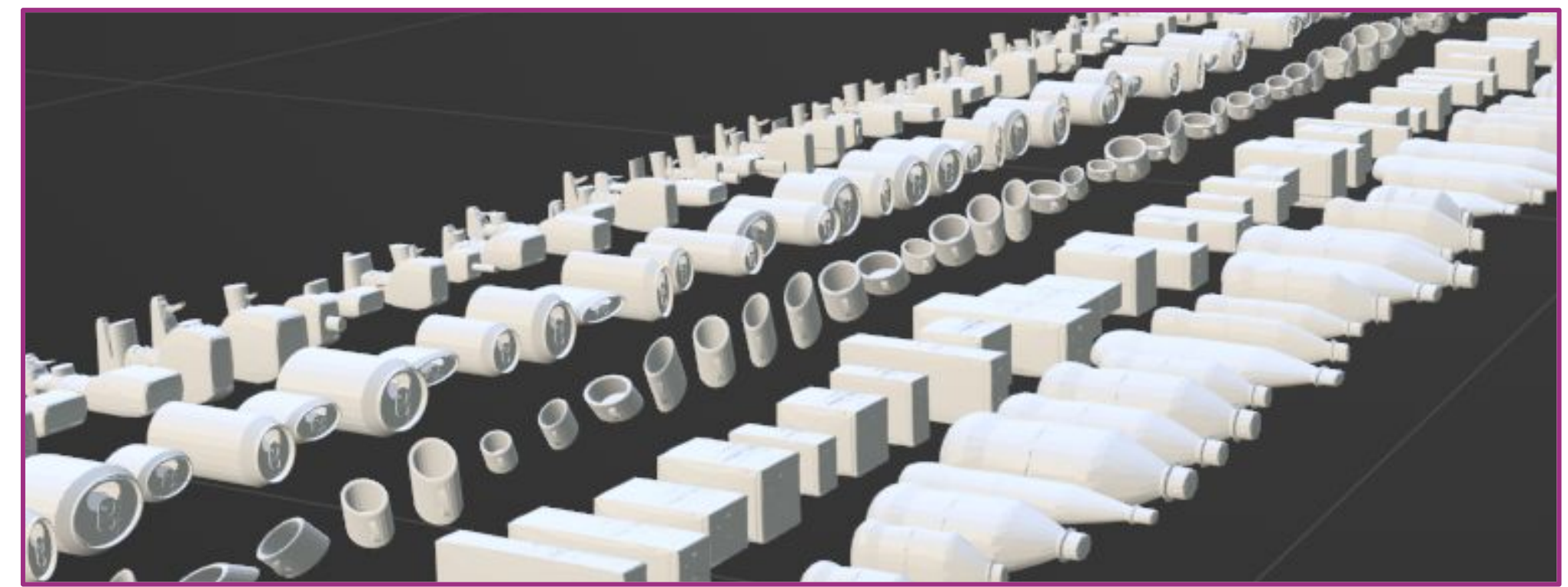
- Data augmentation = point cloud deformation

$$\mathcal{P}_i^a = \{\varphi_{D_a}(p_j) = D_a \cdot p_j \mid p_j \in \mathcal{P}_i\}$$

where $\varphi_{D_a} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$\begin{pmatrix} x_j \\ y_j \\ z_j \end{pmatrix} \mapsto \begin{pmatrix} \alpha_1 & 0 & 0 \\ 0 & \alpha_2 & 0 \\ 0 & 0 & \alpha_3 \end{pmatrix} \begin{pmatrix} x_j \\ y_j \\ z_j \end{pmatrix} \varphi_{D_a}$$

- Final dataset size = 40 353 objects

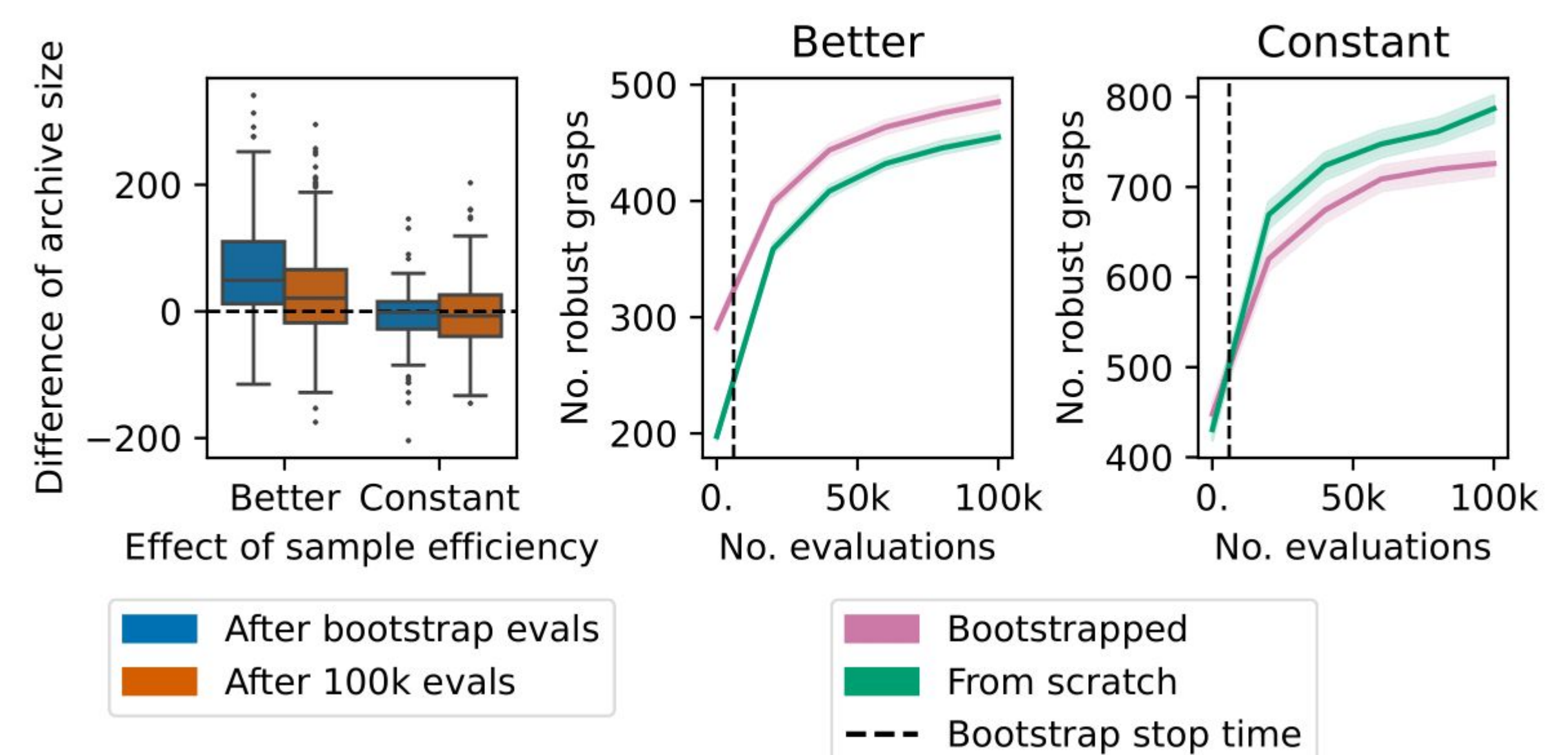


Results

Results of bootstrap on augmented data



- Bootstrap reduces the required number of evaluations by up to 20 %



- Bootstrap preserves grasp quality

References

- [1] Huber, J., Hélénou, F., Kappel, M., Chelly, E., Khoramshahi, M., Amar, F. B. & Doncieux, S. (2024). Speeding up 6-DoF Grasp Sampling with Quality-Diversity. (IROS'24)
- [2] Huber, J., Hélénou, F., Watrelot, H., Amar, F. B. & Doncieux, S. (2024). Domain randomization for sim2real transfer of automatically generated grasping datasets

